

Question 1 (10 marks)

- a) Evaluate $|-2| - |-4|$ (1)
- b) Write $49^{-\frac{3}{2}}$ as a simple fraction (1)
- c) Find $\cot 102^\circ 13'$ correct to 3 decimal places. (2)
- d) Factorise fully $x^4 - 4x^2$ (2)
- e) Solve $5 = \frac{2}{5}(w + 4)$ (2)
- f) Find the values of a and b if $(2 + \sqrt{3})^2 = a + \sqrt{b}$ (2)

Question 2 (10 marks)

- a) For $f(x) = \frac{2}{x+1}$
- i) Write down the domain of the function (1)
- ii) Find $f(\frac{1}{a})$ as a simple fraction (2)
- b) Solve $x^2 = 2x$ (1)
- c) Solve $\sin \theta = \frac{-\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$ (2)
- d) i) Simplify $(2x + h)^2 - 4x^2$ (1)
- ii) Hence evaluate $\lim_{h \rightarrow 0} \frac{(2x + h)^2 - 4x^2}{h}$ (1)
- e) Solve $\frac{|x|}{2} < 1$ (2)

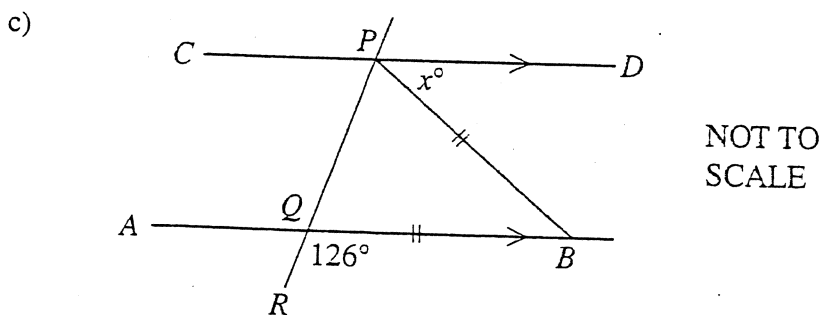
Question 3 (10 marks)

- a) Differentiate the following
- i) $y = \frac{1}{3}x^4 + k$ (1)
- ii) $y = \frac{4x^3 + x^4}{x^2}$ (2)

- b) The points $(2, 7)$ and $(-4, -5)$ are the end points of the diameter of a circle
- Find the coordinates of the centre of the circle (1)
 - Find the length of the radius (2)
- c) If $x^2 + 2x + m = 0$ has roots α and β
- Without finding the roots, find the value of
 - $\alpha + \beta$ (1)
 - $\alpha\beta$ (1)
 - If $\beta = 2\alpha$. Find the value of m (2)

Question 4 (10 marks)

- a) Find the equation of the tangent to the curve $y = 3x^2 + x$ at $x = 1$ (2)
- b)
 - Explain why the lines $y = 2x - 1$ and $6x - 3y + 5 = 0$ are parallel. (2)
 - If $(a, 5)$ lies on $y = 2x - 1$, find the value of a (1)
 - Hence find the distance between the parallel lines in part (i) (2)



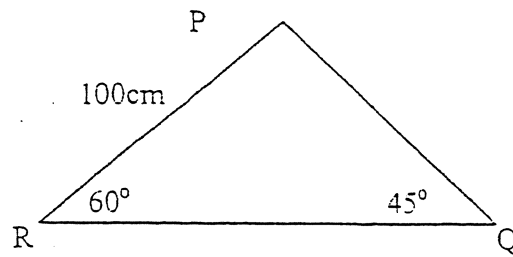
In the diagram CD is parallel to AB and $PB = QB$.

- Find the size of $\angle PQB$ in degrees (1)
- Find the value of x giving reasons (2)

Question 5 (10 marks)

- a)
 - Write $\sqrt[3]{x}$ in index form (1)
 - If $f(x) = \sqrt[3]{x}$. Find $f'(8)$ as a fraction (2)

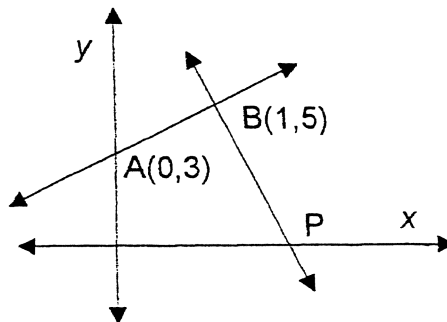
- b) For the parabola $y = x^2 - 6x + 4$
- Find the coordinates of the vertex (2)
 - Sketch the parabola showing the vertex and y – intercept (1)
 - Use your graph or otherwise determine the smallest value of k so that $x^2 - 6x + k$ is positive for all values of x (2)
- c) In ΔPQR (2)



Show that the length of PQ is $50\sqrt{6}cm$

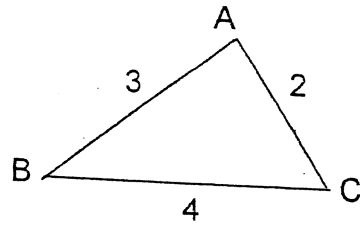
Question 6 (10 marks)

- a) If $y = (x^2 - 5)^5$. Find $\frac{dy}{dx}$ (2)
- b)



- Find gradient of line AB (1)
- If PB is perpendicular to AB find the equation of PB in general form. (3)
- Find the coordinates of P (1)
- If ABPQ form a rectangle find the coordinates of Q (1)

c)



- i) Use the cosine rule to find the exact value of $\cos A$ (2)
- ii) Hence find the exact value of $\sin A$ (1)

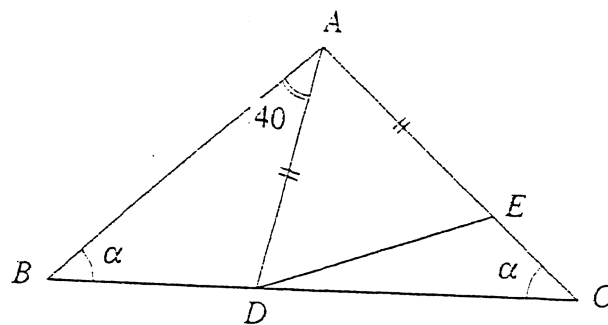
Question 7 (10 marks)

- a)
 - i) Write down the discriminant of $x^2 + px + (p+3)$ (1)
 - ii) If the equation $x^2 + px + (p+3) = 0$ has equal roots find the values of p . (2)

b) If $y = \frac{x}{x^2 + 1}$

- i) Find $\frac{dy}{dx}$ (2)
- ii) Find the x values of the points where $\frac{dy}{dx} = 0$ (1)

c)

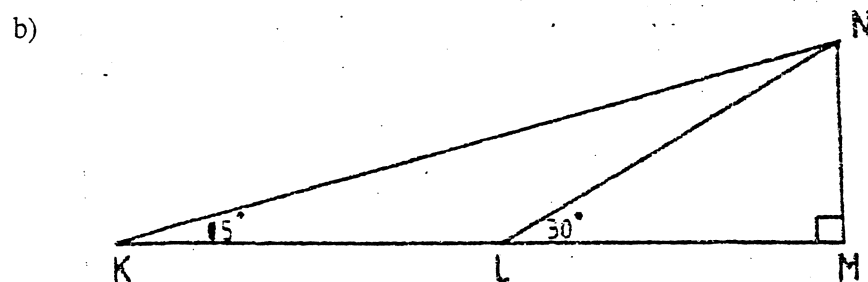


In the isosceles triangle ABC $\angle ABC = \angle ACB = \alpha$ $AD = AE$

- i) Explain why $\angle ADC = \alpha + 40$ (1)
- ii) Find $\angle DAC$ in terms of α (1)
- iii) Hence or otherwise find $\angle EDC$ giving reasons (2)

Question 8 (10 marks)

- a) i) Sketch the curve $y = \frac{8}{x}$ (1)
- ii) Find $\frac{dy}{dx}$ (1)
- iii) Find the equation of the normal to $y = \frac{8}{x}$ at $(4,2)$ (2)
- iv) The normal cuts the curve again at P . Find the coordinates of P . (2)



- i) Explain why $KL = LN$ (1)
- ii) If $NM = 1$ deduce that $\tan 15^\circ = 2 - \sqrt{3}$ (3).

YEAR 11 YEARLY - 2003 - 2 UNIT

Question 1

a) $2 - 4 = -2$

b) $49^{-3/2} = \frac{1}{49^{3/2}}$
 $= \frac{1}{343}$

c) $-0.2165 \dots = -0.217$

d) $x^4 - 4x^2 = x^2(x^2 - 4)$
 $= x^2(x-2)(x+2)$

e) $5 = \frac{2}{5}(w+4)$
 $25 = 2w + 8$
 $17 = 2w$
 $w = 8\frac{1}{2}$

f) $(2+\sqrt{3})^2 = 4 + 4\sqrt{3} + 3$
 $= 7 + 4\sqrt{3}$
 $\therefore a = 7, b = 48$

Question 2

a) i) $x \in \mathbb{R}, x \neq -1$

ii) $\frac{2}{\frac{1}{a} + 1} = \frac{2}{\frac{1+a}{a}}$
 $= \frac{2a}{1+a}$

b) $x^2 - 2x = 0$
 $x(x-2)$
 $x = 0 \text{ or } 2$

c) $\sin \theta = -\frac{\sqrt{3}}{2}$
 $\theta = 60^\circ (3^\text{rd}, 4^\text{th})$
 $\theta = 240^\circ, 300^\circ$

d) i) $(2x+h)^2 - 4x^2$
 $= 4x^2 + 4xh + h^2 - 4x^2$
 $= 4xh + h^2$

ii) $4x$

e) $\frac{|x|}{2} < 1$
 $|x| < 2$
 $-2 < x < 2$

Question 3

a) i) $\frac{dy}{dx} = \frac{4}{3}x^3$

ii) $y = 4x + x^2$
 $\frac{dy}{dx} = 4 + 2x$

b) i) centre = $2\frac{-4}{2}, \frac{7-5}{2}$
 $= (-1, 1)$

ii) radius = $\frac{1}{2}\sqrt{12^2 + 6^2}$
 $= \frac{1}{2}\sqrt{180}$
 $= 3\sqrt{5}$

c) i) -2

ii) m

iii) $3x = -2$
 $x = -2/3$
 $2x^2 = m$
 $\therefore 2 \times 4/9 = m$
 $m = 8/9$

Question 4

a) $y = 3x^2 + x$

$$\frac{dy}{dx} = 6x + 1$$

at $x=1$ $y=4$, $\frac{dy}{dx} = 7$

$$\therefore y - 4 = 7(x - 1)$$

$$y = 7x - 3$$

b) i) $y = 2x - 1$ $m=2$

$$3y = 6x + 5 \quad m=2$$

they have the same gradient ($m_1 = m_2 = 2$)

ii) $5 = 2a - 1$

$$a = 3$$

iii) $d = \left| \frac{3 \times 6 - 3 \times 5 + 5}{\sqrt{36 + 9}} \right|$

$$= \frac{8}{\sqrt{45}}$$

c) i) $\angle PQB = 54^\circ$

ii) $\angle QPB = 54^\circ$ (base $\angle$ of isos Δ)

$\angle QPD = 126^\circ$ (corresponding $\angle$ on $\parallel$ lines)

$$\therefore x + 54^\circ = 126$$

$$x = 72^\circ$$

Question 5

a) i) $x^{1/3}$

ii) $f'(x) = \frac{1}{3}x^{-2/3}$

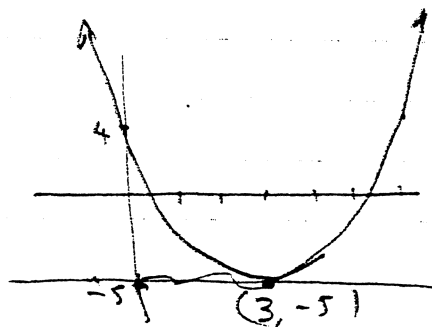
$$f'(8) = \frac{1}{3} \times \frac{1}{8^{2/3}} = \frac{1}{12}$$

b) $y + 4 + 9 = x^2 - 6x + 9$

$$(y + 5) = (x - 3)^2$$

$\therefore$ Vertex $(3, -5)$

ii)



iii) need to add 5

add $\therefore$ y-intercept $4 + 5 = 9$

$$\therefore x^2 - 6x + 9$$

$$\therefore h > 9$$

c) $\frac{PQ}{\sin 60^\circ} = \frac{100}{\sin 45^\circ}$

$$PQ = \frac{100 \sin 60^\circ}{\sin 45^\circ}$$

$$= \frac{100 \times \frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}}$$

$$= 50\sqrt{3} \times \sqrt{2}$$

$$= 50\sqrt{6}$$

Question 6

a) $y = (x^2 - 5)^5$

$$\frac{dy}{dx} = 5 \cdot 2x \cdot (x^2 - 5)^4$$

$$= 10x(x^2 - 5)^4$$

b) i) $m_{AB} = \frac{5 - 3}{1 - 0}$

$$= 2$$

ii) $m_L = -1/2$

$$y - 5 = -1/2(x - 1)$$

$$2y - 10 = -x + 1$$

$$x + 2y - 11 = 0$$

iii) $y = 0$

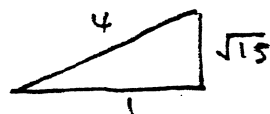
$$x = 11 \quad P(11, 0)$$

iv) $Q(10, -2)$

Q6

$$\begin{aligned} \text{i) } \cos A &= \frac{3^2 + 2^2 - 4^2}{2 \times 3 \times 2} \\ &= -\frac{3}{12} \quad \left(-\frac{1}{4}\right) \end{aligned}$$

ii)



$$\sin A = \frac{\sqrt{15}}{4}$$

Question 7

$$\begin{aligned} \text{a) i) } \Delta &= p^2 - 4(p+3) \\ &= p^2 - 4p - 12 \end{aligned}$$

$$\begin{aligned} \text{ii) } p^2 - 4p - 12 &= 0 \\ (p-6)(p+2) &= 0 \\ p &= 6 \text{ or } -2 \end{aligned}$$

$$\begin{aligned} \text{b) i) } \frac{dy}{dx} &= \frac{(x^2+1) \cdot 1 - x \cdot 2x}{(x^2+1)^2} \\ &= \frac{1-x^2}{(x^2+1)^2} \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{1-x^2}{(x^2+1)^2} &= 0 \\ \therefore x &= \pm 1 \end{aligned}$$

c) i) exterior $\angle$ of triangle

$$\begin{aligned} \text{ii) } x + x + 40 + \angle DAC &= 180^\circ \\ \angle DAC &= 140 - 2x \end{aligned}$$

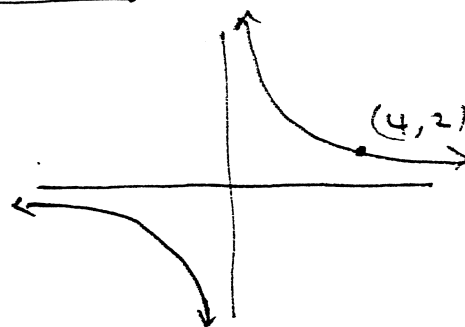
iii) Now $\angle ADE = \angle DEA$ (base angles of isosceles Δ) y .

$$\begin{aligned} \therefore 2y + 140 - 2x &= 180^\circ \\ y &= 20 + x. \end{aligned}$$

$$\begin{aligned} \angle EDC &= \angle ADC - \angle ADE \\ &= x + 40 - (20 + x) \\ &= 20 \end{aligned}$$

Question 8

i)



$$\begin{aligned} \text{ii) } y &= 8x^{-1} \\ \frac{dy}{dx} &= -8x^{-2} \\ &= -\frac{8}{x^2} \end{aligned}$$

$$\text{iii) at } (4, 2) \quad \frac{dy}{dx} = -\frac{1}{2}$$

$$\therefore m_{\perp} = 2$$

$$y - 2 = 2(x - 4)$$

$$y = 2x - 6$$

$$\text{iv) } \therefore \frac{8}{x} = 2x - 6$$

$$8 = 2x^2 - 6x$$

$$2x^2 - 6x - 8 = 0$$

$$2(x-4)(x+1) = 0$$

$$\therefore x = 4 \text{ or } -1$$

$$\therefore P(-1, -8)$$

b) i) $\angle KNL = 15^\circ$ (exterior $\angle$ of Δ)

$$\therefore KL = LN$$

$$\begin{aligned} \text{ii) } \sin 30^\circ &= \frac{1}{NL} & \cos 30^\circ &= \frac{LM}{2} \\ \therefore NL &= 2 & LM &= \sqrt{3} \\ \therefore KL &= 2 \end{aligned}$$

In ΔKNM

$$\begin{aligned} \tan 15^\circ &= \frac{1}{KL + LM} \\ &= \frac{1}{2 + \sqrt{3}} \\ &= \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}} \\ &= 2 - \sqrt{3} \end{aligned}$$